Diagnosis: Non-small cell lung cancer

Treatement 1: Afatinib

Chosen clinical trials:

1. 'Afatinib in EGFR+NSCLC (Recurrent or Stage IV) - Patients With Poor Performance Status (ECOG 2 or 3)' terminated phase4 https://clinicaltrials.gov/study/NCT02695290

Description: Percentage of patients with occurrence of Common Terminology Criteria for Adverse Events (CTCAE) grade 3 or higher diarrhoea, rash/acne+, stomatitis+ and paronychia+ (+ represents grouped term)

Population: 1

Time: Up to 98 days

group description: Afatinib

measure: Percentage of patients with occurrence of CTCAE grade 3 or higher diarrhoea, rash/acne+, stomatitis+ and paronychia+ (0.0)

1. 'LUX-Lung 4: BIBW 2992 (Afatinib) Phase I Trial in Advanced Non Small Cell Lung Cancer Patients & Phase II Trial in Non Small Cell Lung Cancer Patients Failing Erlotinib or Gefitinib' completed phase2 https://clinicaltrials.gov/study/NCT00711594

Description: The primary endpoint was the objective response rate (ORR) as assessed by RECIST 1.1 criteria.

Population: 62

Time: Screening visit

group description: BIBW 50mg (Phase II)

measure: Objective response rate (ORR) as assessed by RECIST 1.1 criteria (28.333333333333336)

measure: Duration of response (DoR) in months (10.5)

measure: Progression-free survival (PFS) in months (4.1)

measure: Overall survival (OS) in months (12.3)

Description: Secondary endpoints included safety and tolerability of BIBW 2992.

Population: 62

Time: Screening visit

group description: BIBW 50mg (Phase II)

measure: Adverse events (AEs) reported (56.0)

measure: Grade 3 or higher AEs (14.0)

measure: Treatment-related AEs (32.0)

measure: Dose reduction due to AEs (10.0)

measure: Discontinuation due to AEs (4.0)

Description: The secondary endpoint included the summary of EGFR mutation findings.

Population: 62

Time: Screening visit

group description: BIBW 50mg (Phase II)

measure: EGFR Mutation test done (local or central) (56.0)

measure: Positive (45.0)

measure: Del19 (22.0)

measure: Del19 + L858R (1.0)

measure: Del19 + T790M (1.0)

measure: Del19 + Other (1.0)

measure: L858R (15.0)

measure: L858R + T790M (1.0)

measure: L858R + Other (3.0)

measure: L861Q (1.0)

measure: Negative (11.0)

Chosen scientific papers:

Afatinib showed partial reversal of immunosuppressive-like transcriptional signature in resistant NSCLC [1]. Afatinib showed efficacy in older adult EGFR-mutant lung cancer patients [2]. Afatinib is approved for first-line treatment in advanced EGFR mutation-positive NSCLC [3].

[1] EGFR and IRE1α pathways are associated with distinct immunomodulatory gene expression profiles in NSCLC cells with acquired resistance to EGFR TKIs. https://pubmed.ncbi.nlm.nih.gov/41672191

[2] Efficacy and safety findings of the EXTRA study in older adult EGFR-mutant lung cancer patients receiving afatinib as first-line treatment. https://pubmed.ncbi.nlm.nih.gov/41673455

[3] Prospective Observational Real-World Study of Afatinib as First-Line Treatment in Patients with Epidermal Growth Factor Receptor (EGFR) Mutation-Positive Advanced Non-

Small Cell Lung Cancer (NSCLC) in China. https://pubmed.ncbi.nlm.nih.gov/41736995